Directional Derivative

In Euclidean Space

Let

$$f : \mathbb{R}^n \to \mathbb{R}$$

be smooth. The directional derivative of f at x in the direction v is

$$Df(x)[v] = \frac{d}{dt} f(x + tv)\bigg|_{t=0}.$$

It measures the first-order rate of change of f if one starts at x and moves infinitesimally with velocity v.

If f is differentiable, then

$$Df(x)[v] = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(x) v^i.$$

Thus, for fixed x, the map

$$v \mapsto Df(x)[v]$$

is linear in v.

On a Smooth Manifold

Let M be a smooth manifold, let $p \in M$, and let

$$X_p \in T_p M$$

be a tangent vector. A tangent vector can be represented by a smooth curve

$$\gamma : (-\epsilon, \epsilon) \to M$$

such that

$$\gamma(0) = p, \qquad \dot{\gamma}(0) = X_p.$$

The directional derivative of $f : M \to \mathbb{R}$ at p in the direction X_p is

$$X_p[f] = \frac{d}{dt} f(\gamma(t))\bigg|_{t=0}.$$

This number depends only on the tangent vector X_p, not on the particular curve chosen to represent it.

Relation to the Differential

The differential of f at p is the covector

$$df_p : T_p M \to \mathbb{R}$$

defined by

$$df_p(X_p) = X_p[f].$$

So the differential packages all directional derivatives at p into one linear map.

Is It the Covariant Derivative?

For a scalar function, the directional derivative $X[f]$ is already intrinsic. It does not require a connection.

A connection is needed when differentiating vector fields or tensor fields along directions, because values of vector fields at different points live in different tangent spaces. But a function has values in $\mathbb{R}$, so along a curve $\gamma(t)$ we can simply differentiate the real-valued function

$$t \mapsto f(\gamma(t)).$$

Therefore:

$$\text{directional derivative of a function: no connection needed,}$$

while

$$\text{covariant derivative of vector/tensor fields: connection needed.}$$